SISTEMA DE ANALISIS Y REPORTE DE DATOS

Intercambiador de Calor — Preheat Train de Crudo

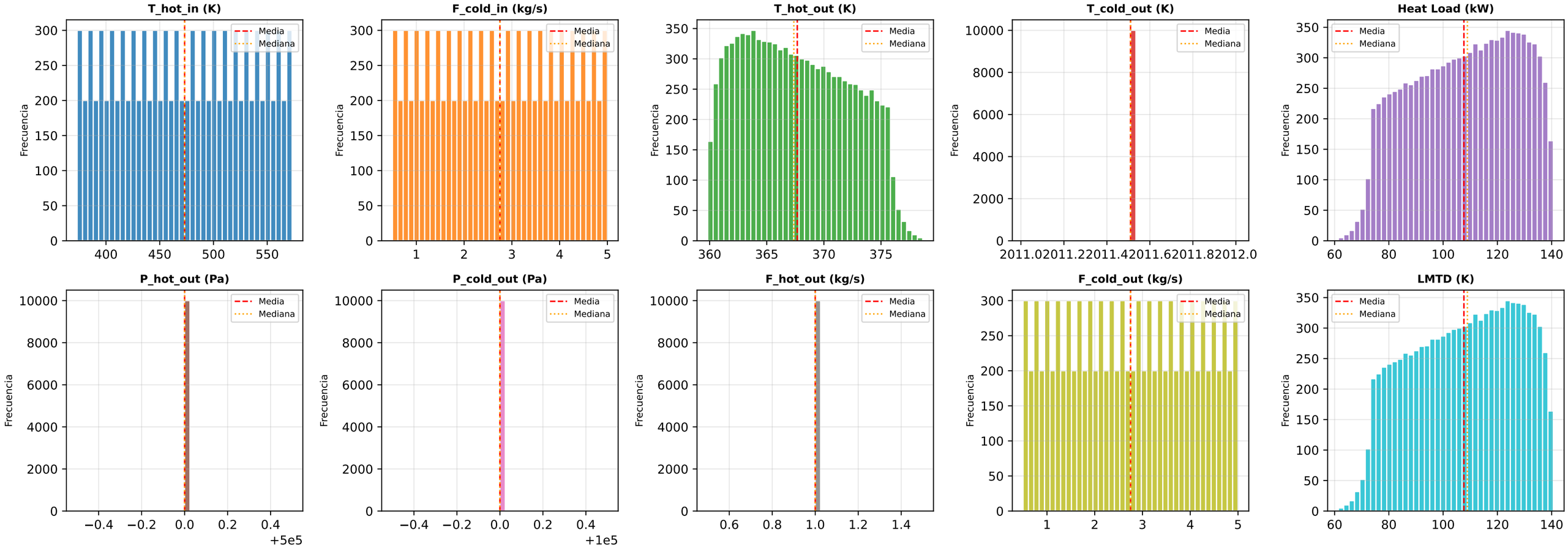
Monitoreo continuo | Deteccion de anomalias | Modelo GRU predictivo

Dataset: Heat Exchanger Parametric Study (DWSIM / Kaggle)

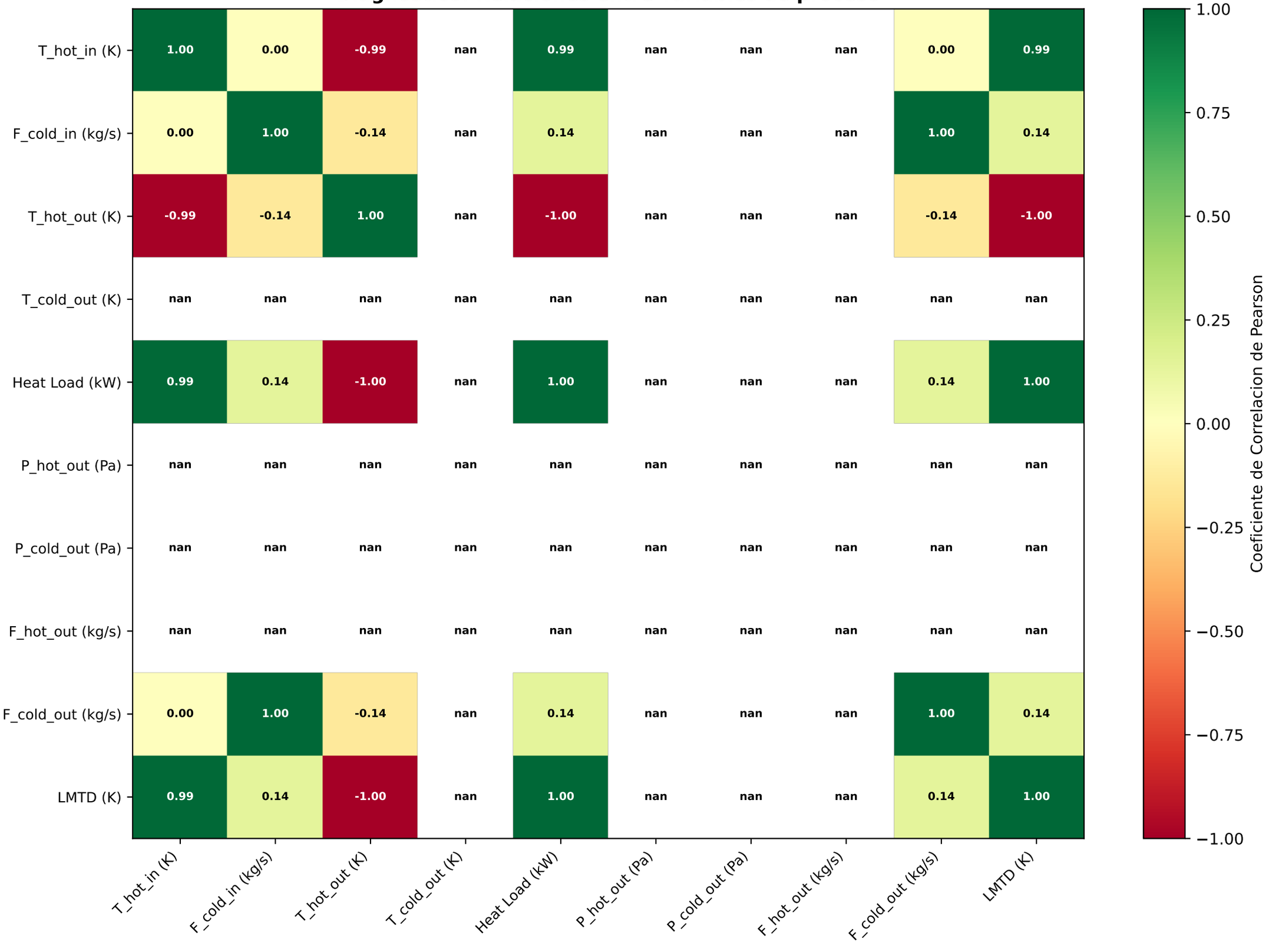
Aplicacion: Industria Petrolera — YPFB Refinacion

Generado: 09/04/2026

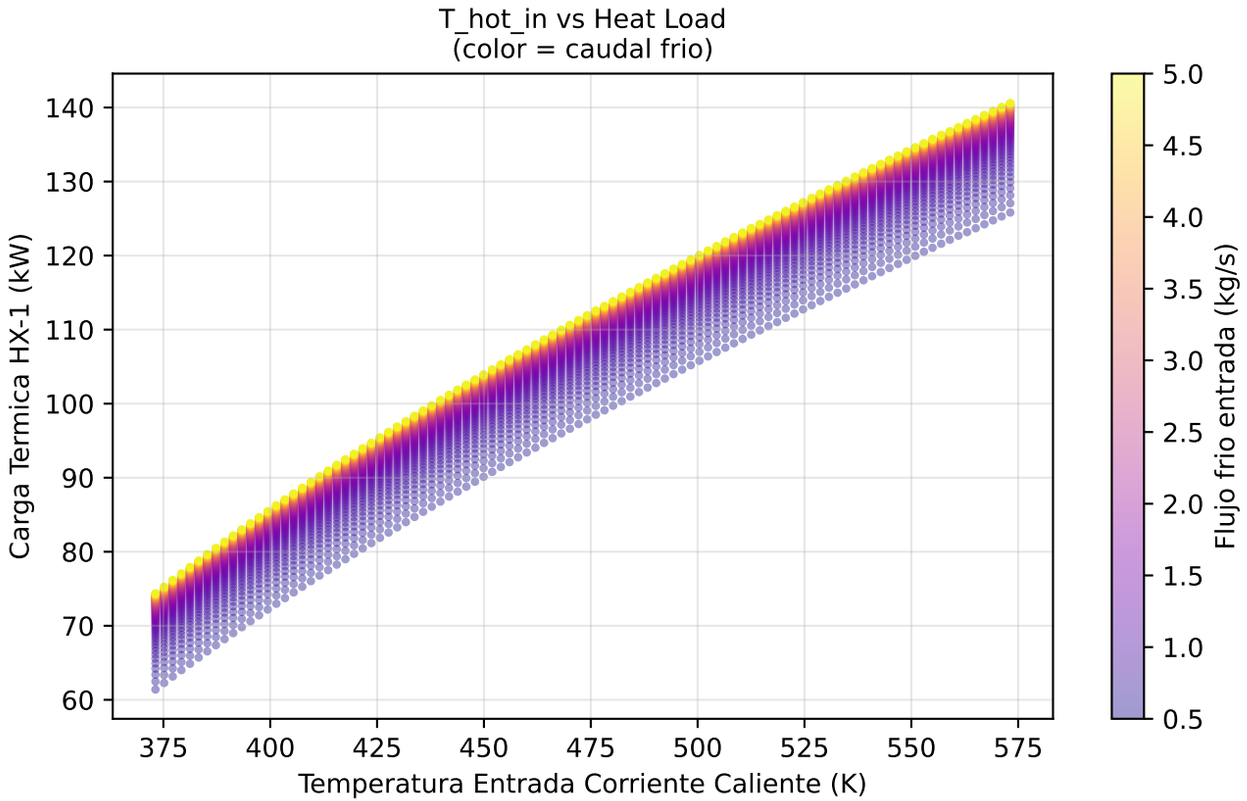
Distribucion de Variables del Proceso — Intercambiador de Calor
Aplicacion: Preheat Train de Crudo | YPFB Refinacion

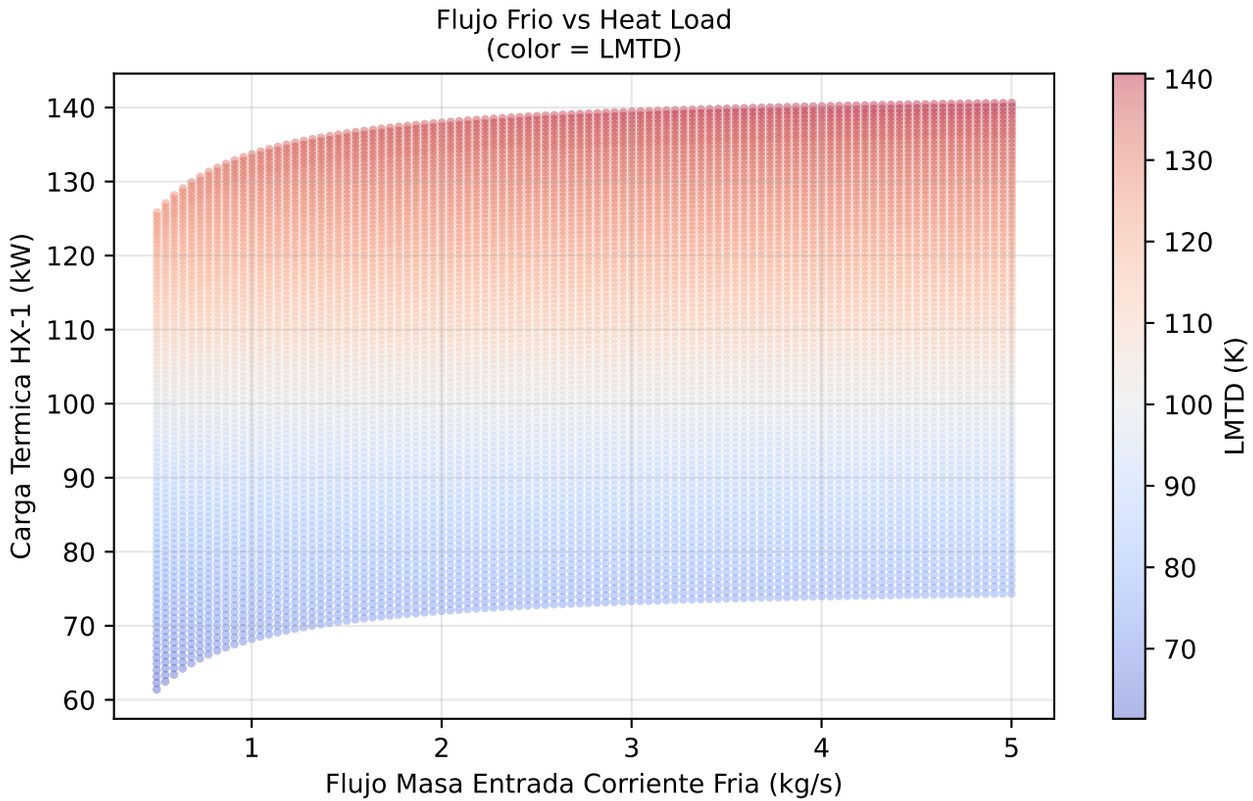


Mapa de Calor de Correlaciones — Variables del Intercambiador
Diagnostico: Relaciones entre variables de proceso

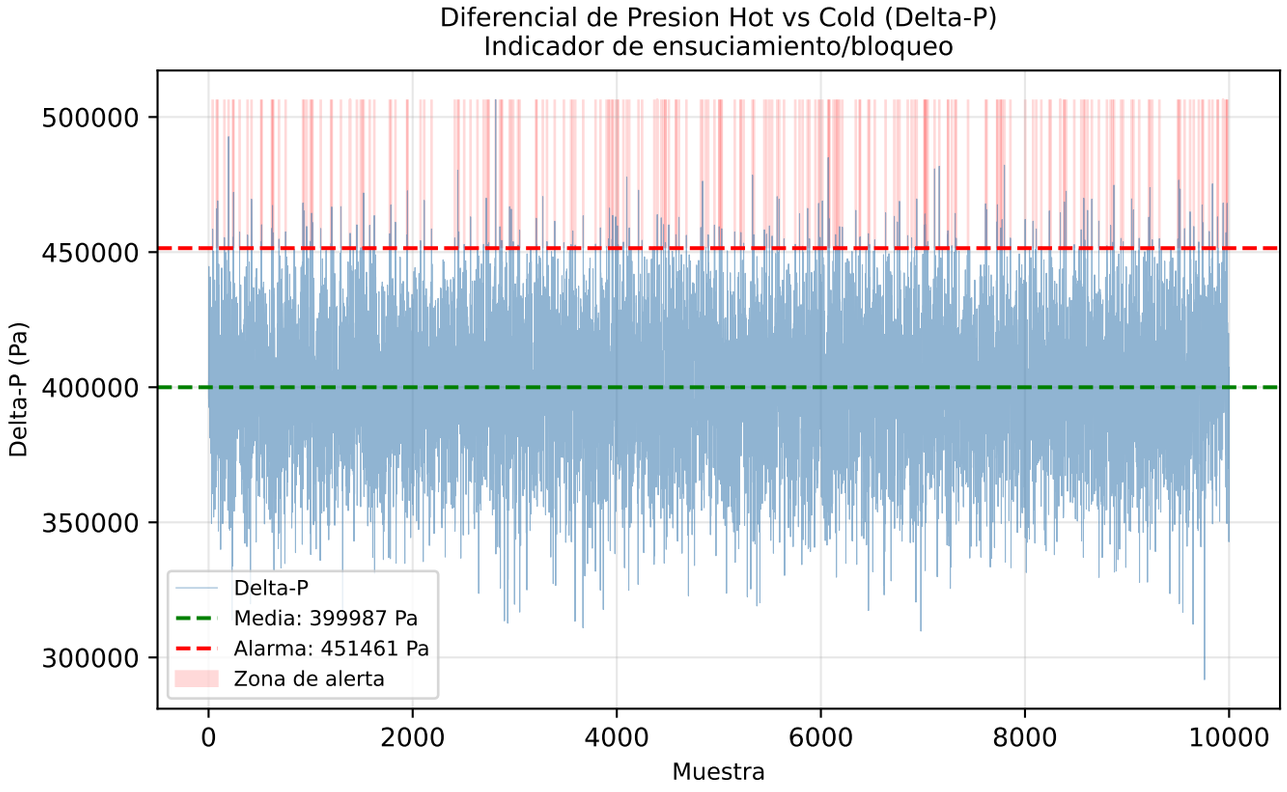


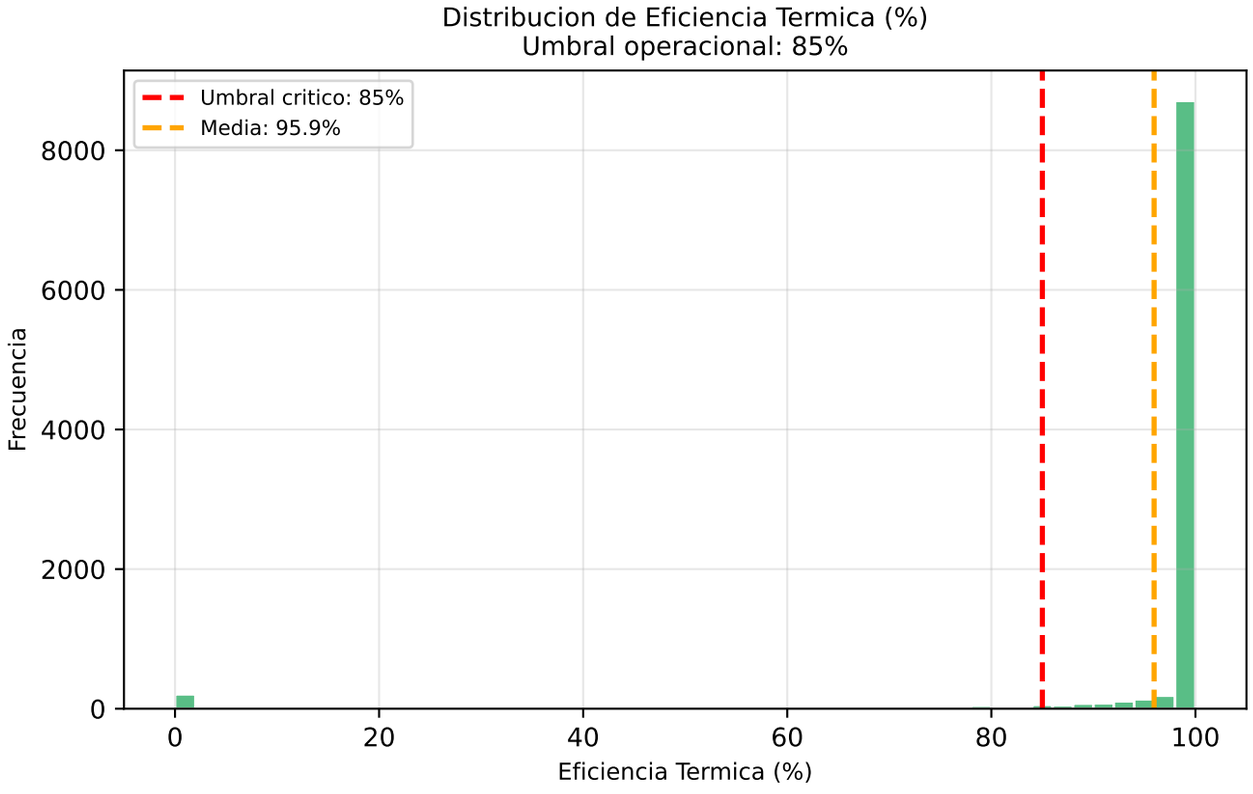
Curva de Operacion del Intercambiador — Diagnostico de Desempeno
Aplicacion: Deteccion de operacion fuera de diseno



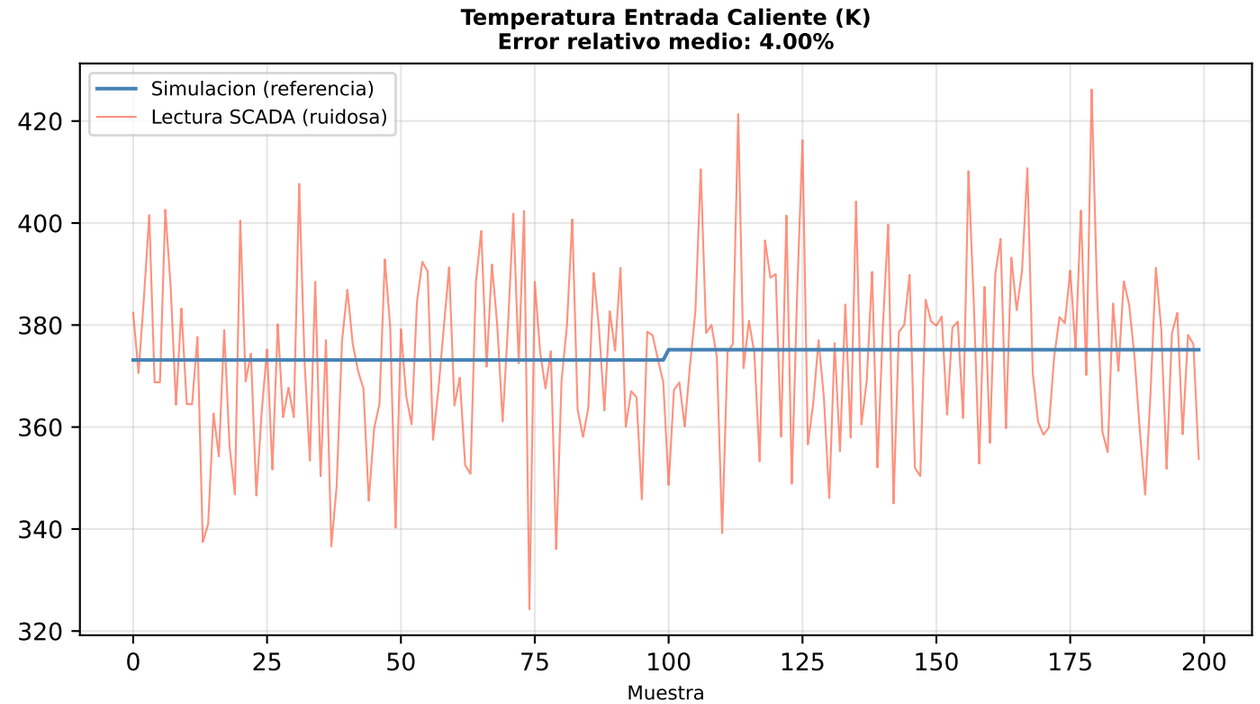


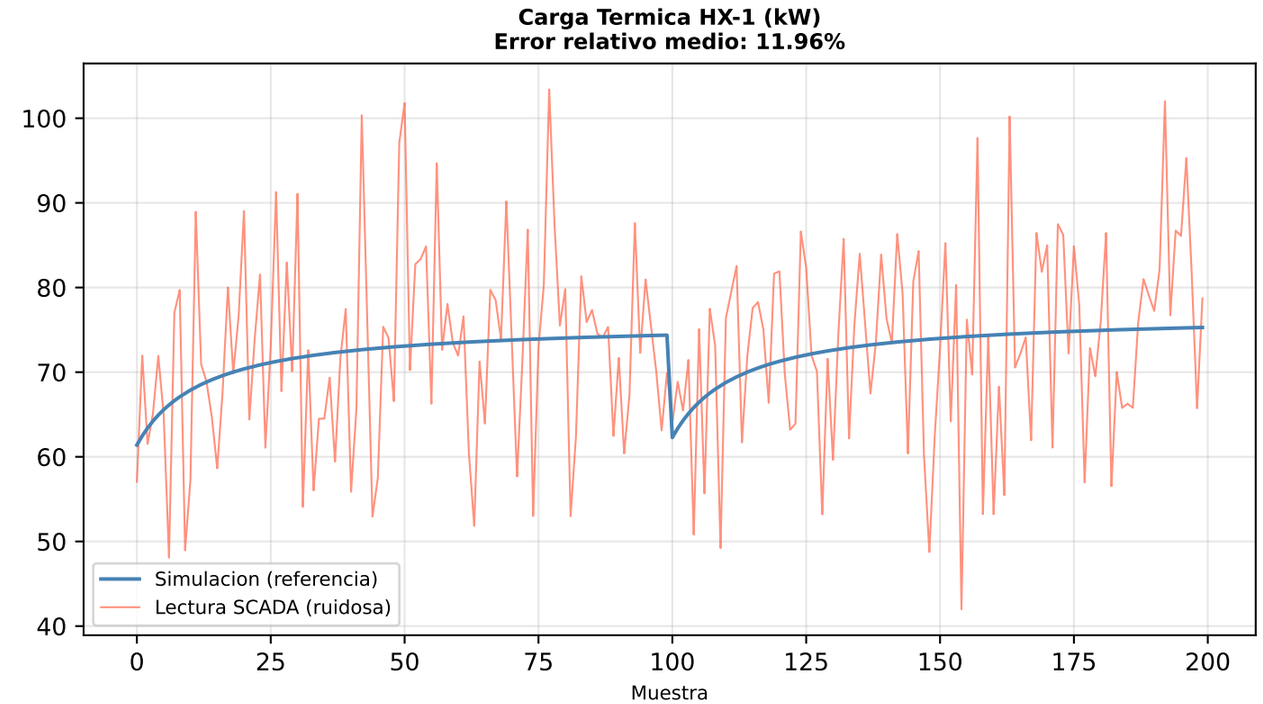
Indicadores Operacionales Clave (KPIs) — Intercambiador de Calor
Aplicacion: Monitoreo continuo en sala de control

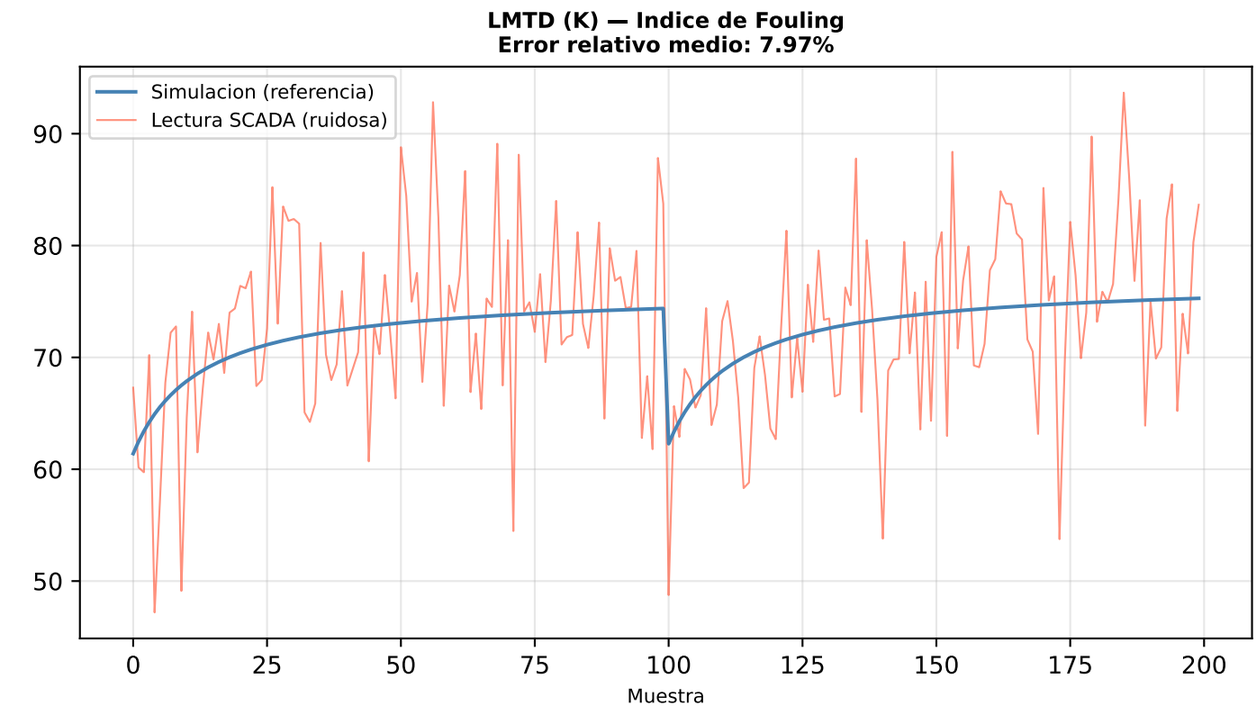


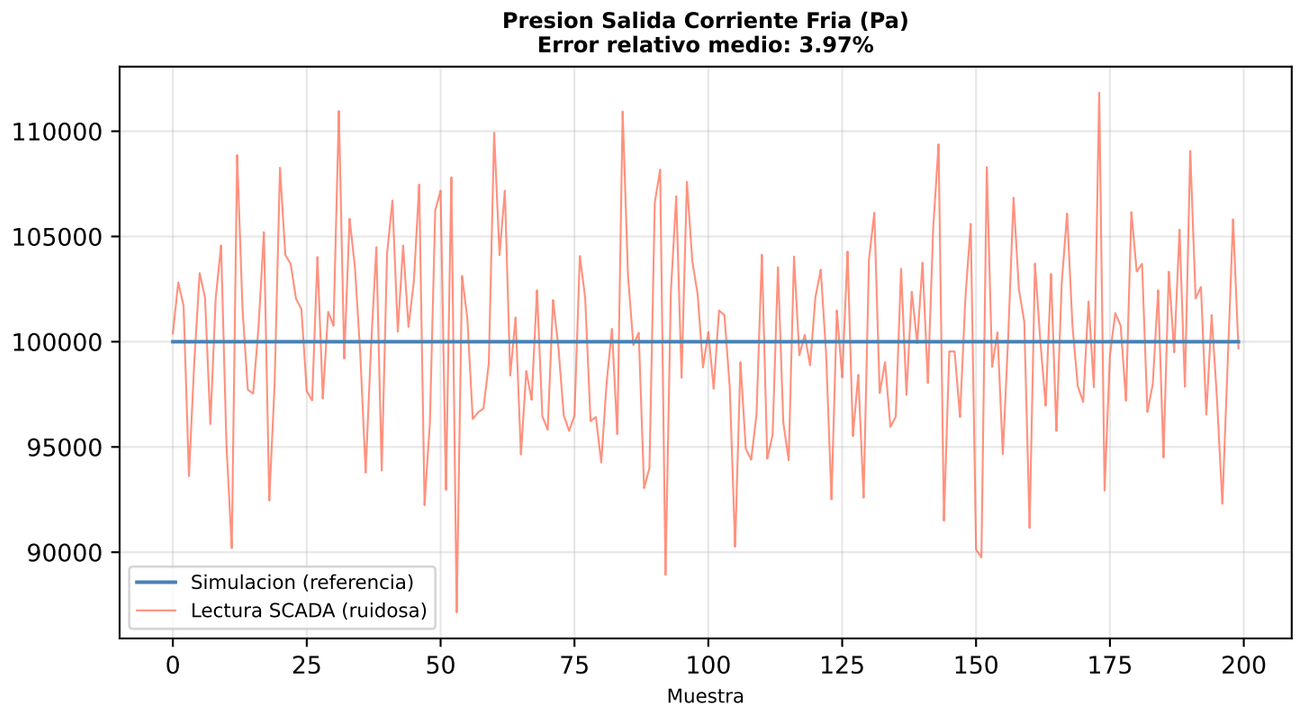


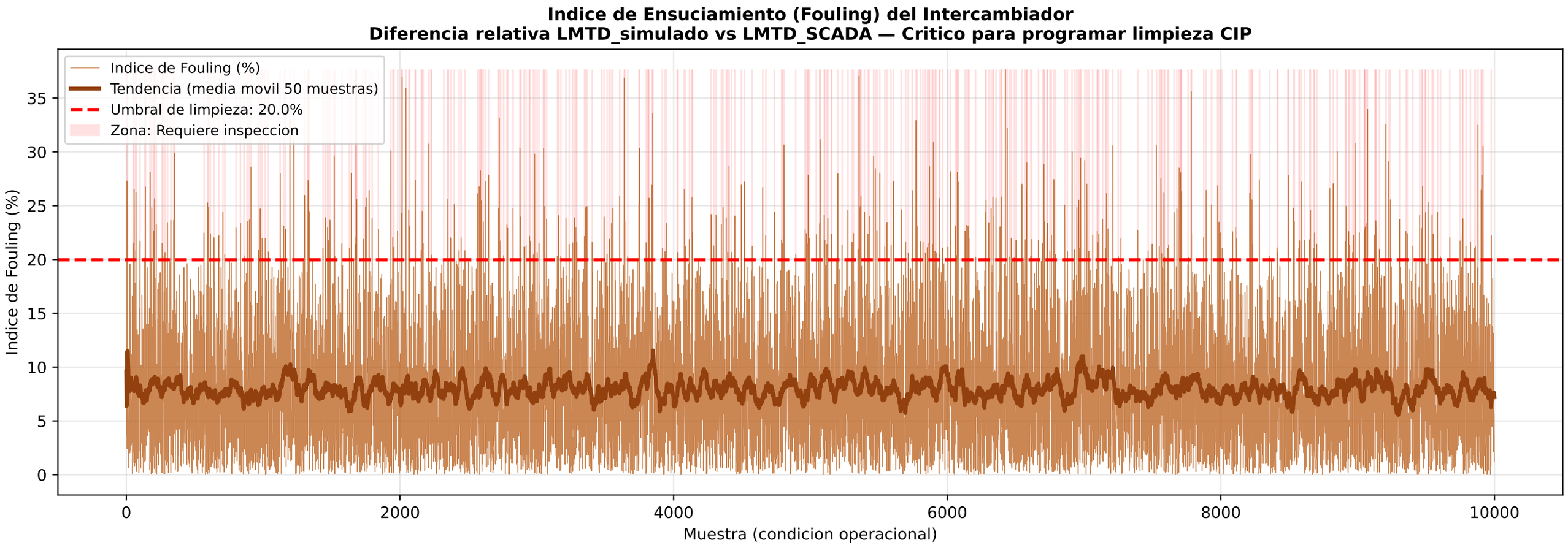
Comparacion Senal de Simulacion vs Lectura SCADA (con ruido gaussiano)
Aplicacion: Calibracion y validacion de instrumentos en campo



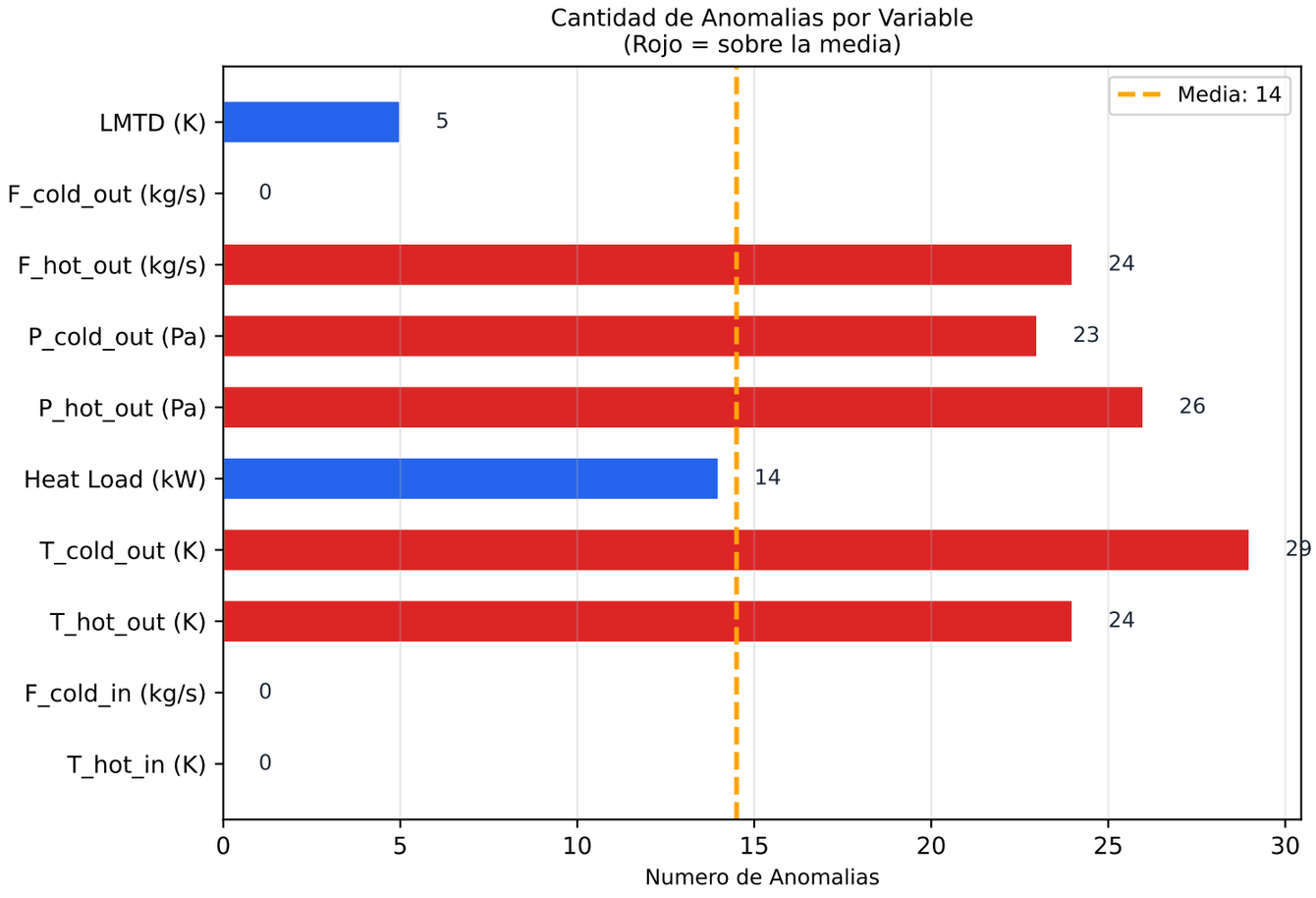


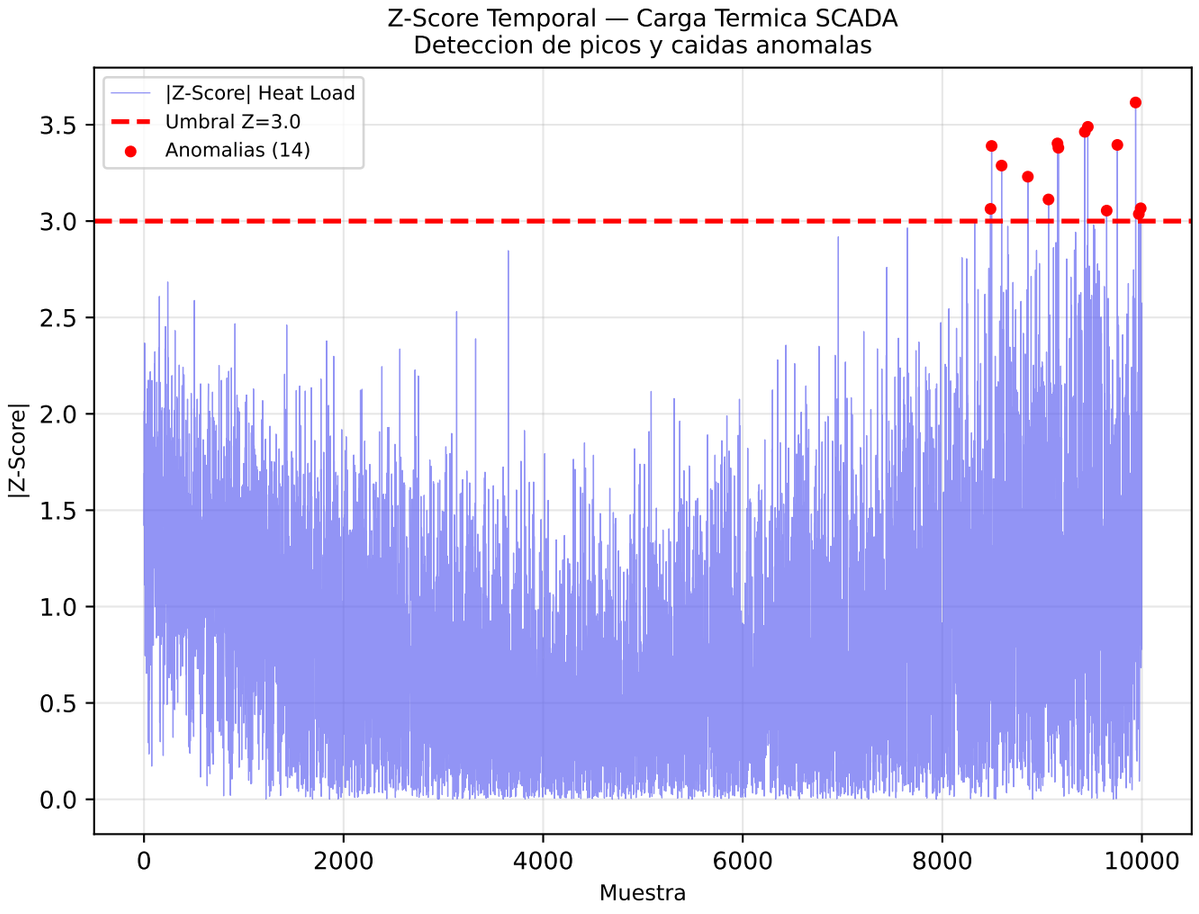


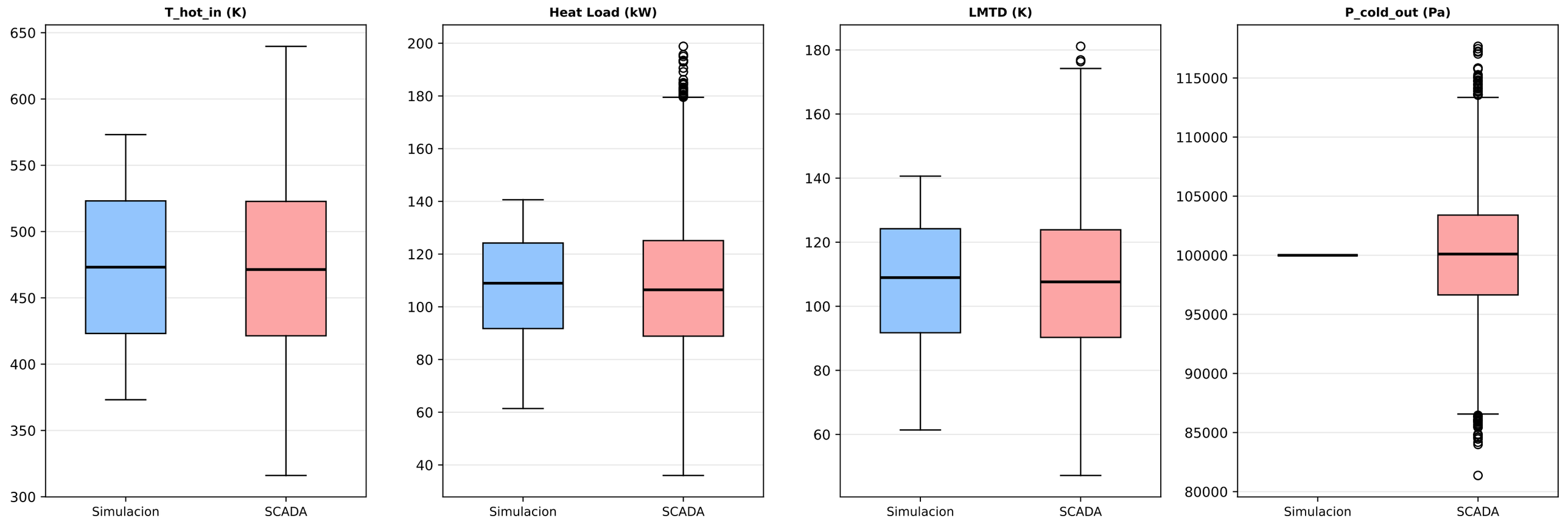



Indice de Ensuciamiento (Fouling) del Intercambiador
Diferencia relativa LMTD_simulado vs LMTD_SCADA — Critico para programar limpieza CIP
Indice de Fouling (%)
Tendencia (media movil 50 muestras)
Umbral de limpieza: 20.0%
Zona: Requiere inspeccion
Indice de Fouling (%)
0
5
10
15
20
25
30
35
0
2000
4000
6000
8000
10000
Muestra (condicion operacional)

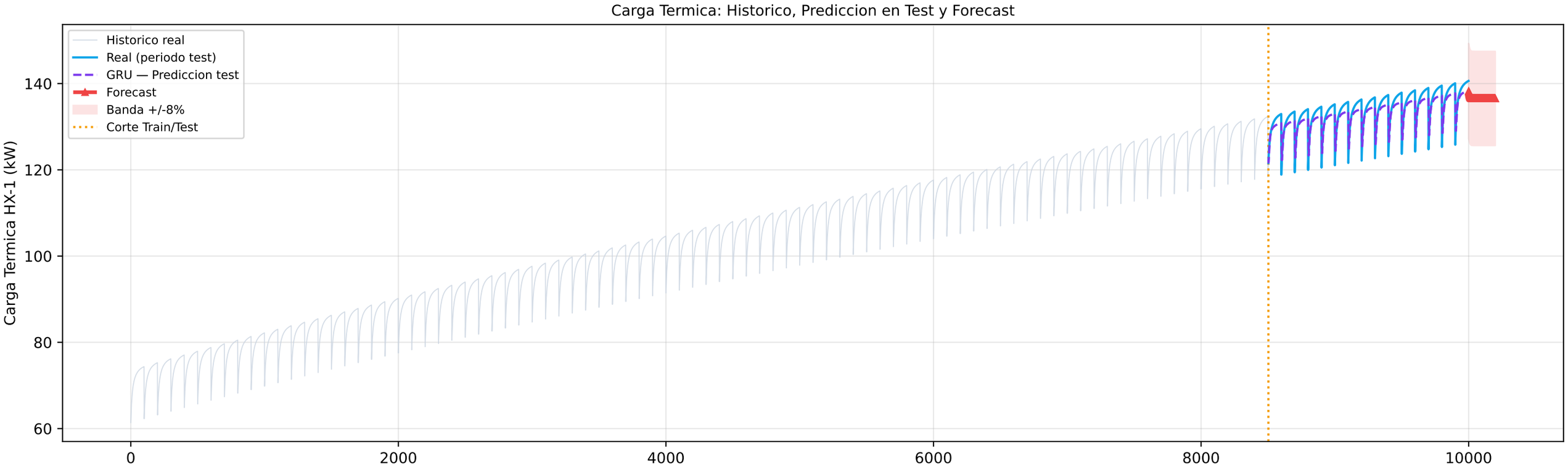
Deteccion de Anomalias en Senales SCADA — Metodo Z-Score (|Z| > 3)
Aplicacion: Alertas de instrumentacion y fallas de sensores

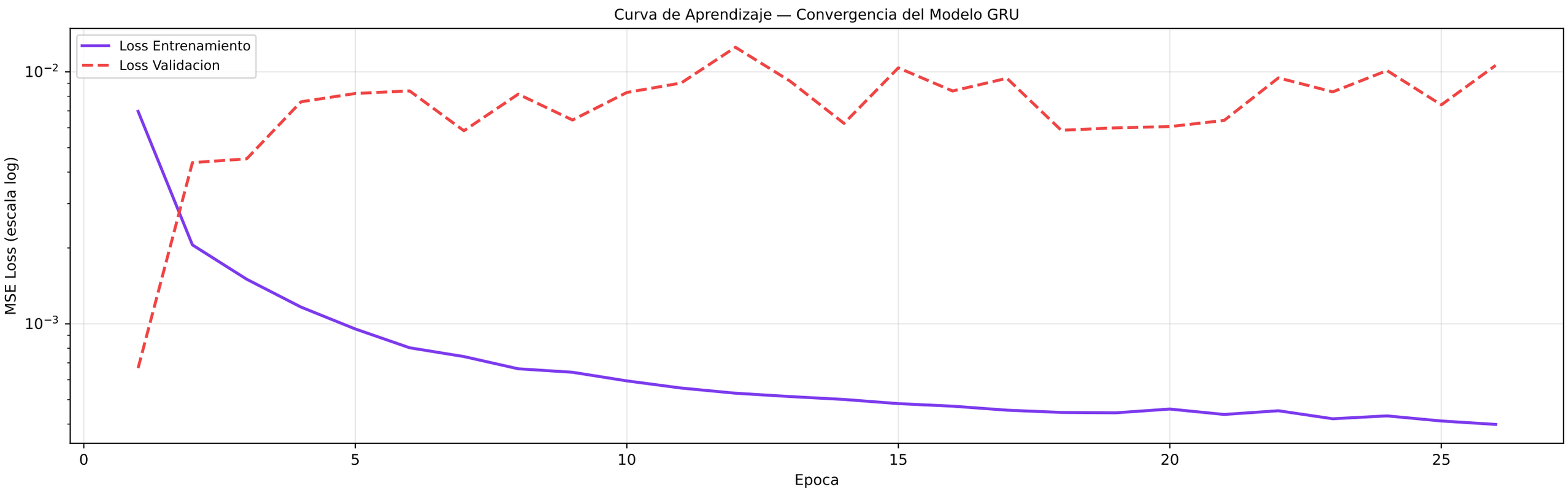
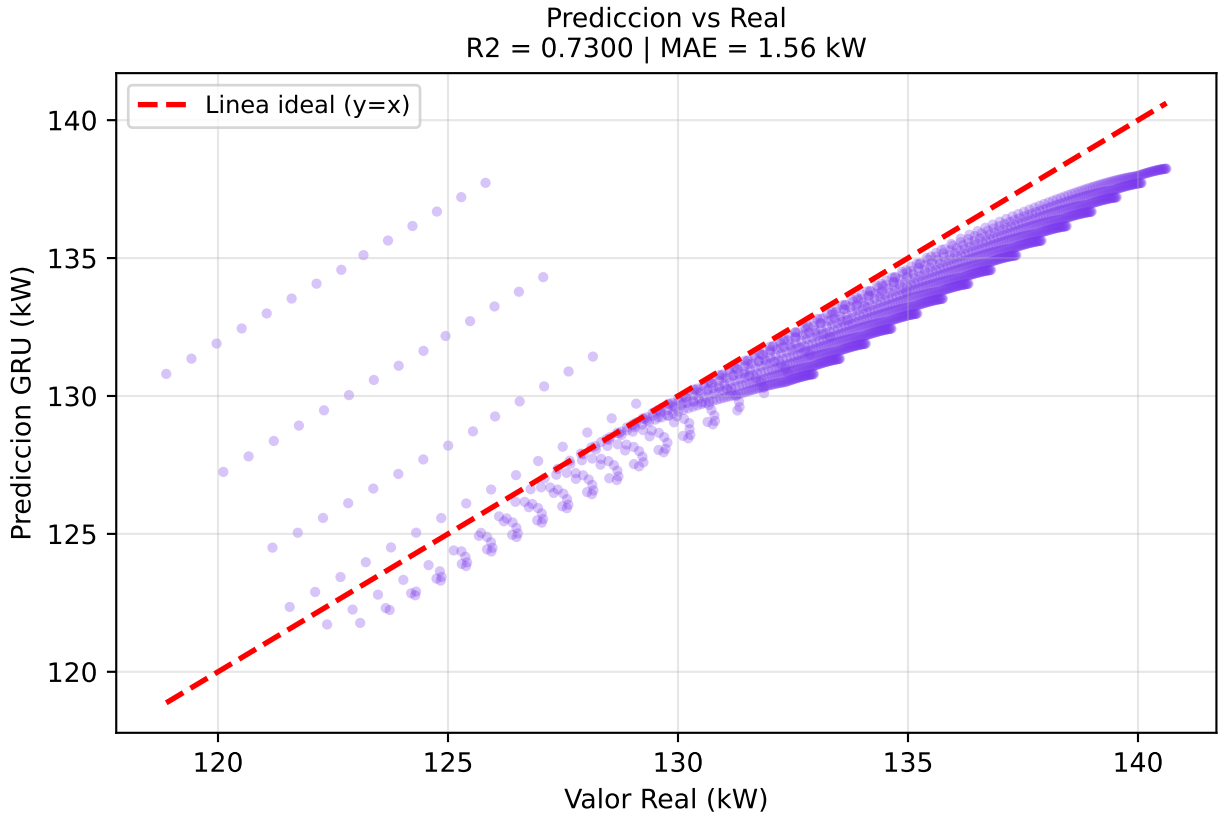
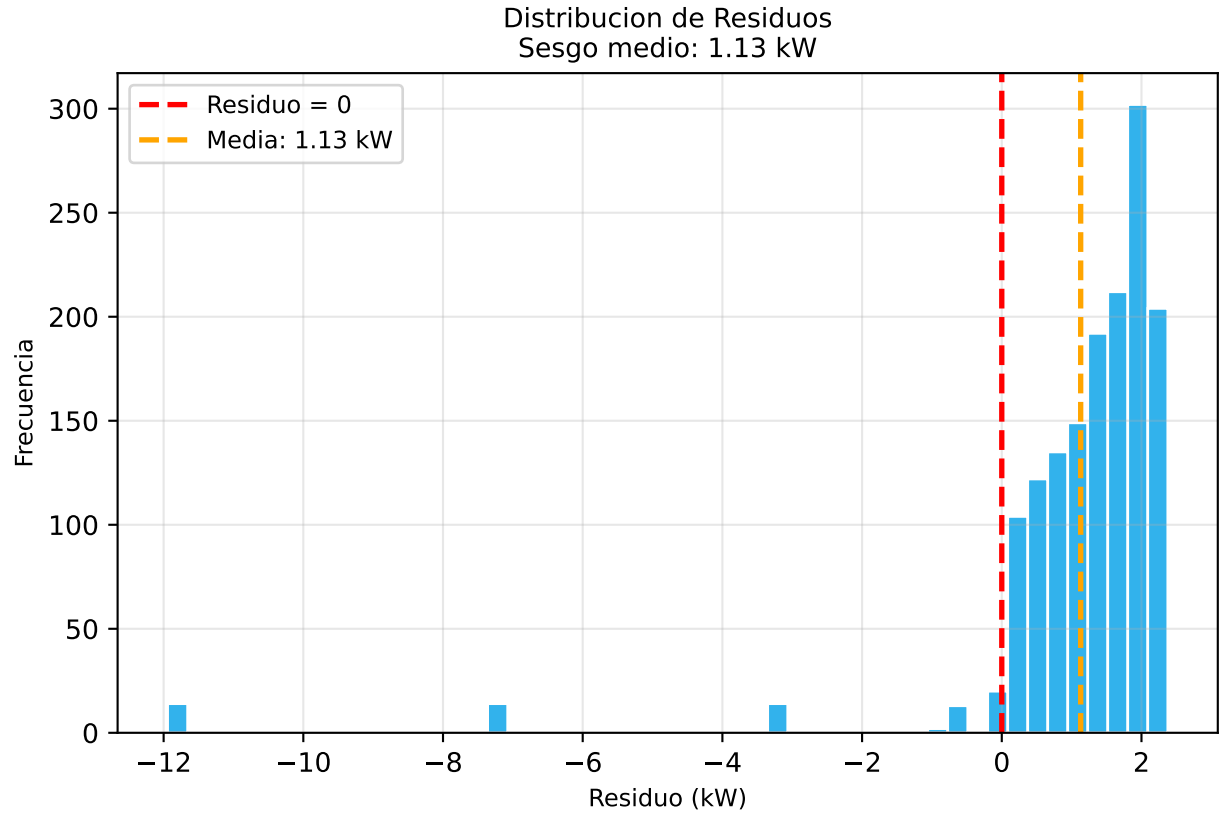





Boxplot: Valores Atipicos — Simulacion vs SCADA
Aplicacion: Validacion de calibracion de instrumentos en campo
T_hot_in (K)
Heat Load (kW)
LMTD (K)
P_cold_out (Pa)
Simulacion
SCADA

GRU — Prediccion y Forecast de Carga Termica HX-1 (kW)
MAPE=1.17% | R2=0.7300 | EXCELENTE
Aplicacion: Mantenimiento predictivo y programacion de limpieza CIP





Calidad del Modelo GRU — Prediccion vs Real y Distribucion de Residuos
Aplicacion: Validacion del modelo antes de despliegue en produccion